

Extremal Shear Stresses

last time $\rightarrow$ extremal normal str

normal stress: $\sigma_n = \underline{n} \cdot \underline{\sigma} \underline{n}$ (quadratic, n_i^2)

shear stress: $|\underline{\tau}|^2 = \sigma_n^2 + \tau^2$ (quartic, n_i^4)

$\Rightarrow$ algebra gets messy.

Mohr circle

$\Rightarrow$ graphical & intuitive

In principal frame we have 3 equations:

1) $\sigma_n = \underline{n} \cdot \underline{\sigma} \underline{n}$ normal stress

2) $|\underline{\tau}|^2 = \tau^2 + \sigma_n^2$ Pythagoras

3) $\underline{n} \cdot \underline{n} = 1$ unit vector

In terms of components:

1) $\sigma_1 n_1^2 + \sigma_2 n_2^2 + \sigma_3 n_3^2 = \sigma_n$

2) $\sigma_1^2 n_1^2 + \sigma_2^2 n_2^2 + \sigma_3^2 n_3^2 = \underbrace{\tau^2 + \sigma_n^2}_q$

3) $n_1^2 + n_2^2 + n_3^2 = 1$

$\Rightarrow$ linear system for $x = n_1^2$ $y = n_2^2$ $z = n_3^2$

Linear system

$$\underbrace{\begin{pmatrix} \sigma_1 & \sigma_2 & \sigma_3 \\ \sigma_1^2 & \sigma_2^2 & \sigma_3^2 \\ 1 & 1 & 1 \end{pmatrix}}_{\underline{\underline{A}}} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \sigma_n \\ q \\ 1 \end{pmatrix}$$
$$\underline{\underline{x}} = \underline{\underline{b}}$$

Solve using Cramers rule:

$$\underline{\underline{A}} \underline{\underline{x}} = \underline{\underline{b}} \quad \det A \neq 0 \quad \underline{\underline{A}} \text{ is } n \times n$$

$$x_k = \frac{\det \underline{\underline{A}}_k}{\det \underline{\underline{A}}} \quad k = 1, \dots, n$$

$\underline{\underline{A}}_k$ is $\underline{\underline{A}}$ with k -th column replaced by $\underline{\underline{b}}$.

$\Rightarrow$ This will give clean answer.

$$\det(\underline{\underline{A}}) = \det(\underline{\underline{A}}^T) = \begin{vmatrix} \sigma_1 & \sigma_1^2 & 1 \\ \sigma_2 & \sigma_2^2 & 1 \\ \sigma_3 & \sigma_3^2 & 1 \end{vmatrix} \xrightarrow{\text{cyclic perm.}} \begin{vmatrix} 1 & \sigma_1 & \sigma_1^2 \\ 1 & \sigma_2 & \sigma_2^2 \\ 1 & \sigma_3 & \sigma_3^2 \end{vmatrix}$$

$\Rightarrow$ Vandermonde matrix $\underline{\underline{A}} = \underline{\underline{V}}(\sigma_1, \sigma_2, \sigma_3)$

$$\det \underline{\underline{V}} = (\sigma_2 - \sigma_1)(\sigma_3 - \sigma_1)(\sigma_3 - \sigma_2)$$

$$\sigma_1 \geq \sigma_2 \geq \sigma_3$$

$$\det \underline{\underline{A}} = -(\sigma_1 - \sigma_2)(\sigma_1 - \sigma_3)(\sigma_2 - \sigma_3)$$

$\Rightarrow$ as long as $\sigma_1, \sigma_2, \sigma_3$ are distinct $\det A \neq 0$

Apply Cramer's rule:

$$\bullet x_1 = n_1^2 = \frac{\det \underline{\underline{A}}_1}{\det \underline{\underline{A}}} \quad \underline{\underline{A}}_1 = \begin{pmatrix} \sigma_n & \sigma_2 & \sigma_3 \\ 0 & \sigma_2^2 & \sigma_3^2 \\ 1 & 1 & 1 \end{pmatrix}$$

$$\det(\underline{\underline{A}}_1) = -(\sigma_2 - \sigma_3) [\tau^2 + (\sigma_n - \sigma_2)(\sigma_n - \sigma_3)]$$

$$\Rightarrow n_1^2 = \frac{\tau^2 + (\sigma_n - \sigma_2)(\sigma_n - \sigma_3)}{(\sigma_1 - \sigma_2)(\sigma_1 - \sigma_3)}$$

$$\bullet x_2 = n_2^2 = \frac{\det \underline{\underline{A}}_2}{\det \underline{\underline{A}}} \quad \underline{\underline{A}}_2 = \begin{pmatrix} \sigma_1 & \sigma_n & \sigma_3 \\ \sigma_1^2 & 0 & \sigma_3^2 \\ 1 & 1 & 1 \end{pmatrix}$$

$$\det(\underline{\underline{A}}_2) = (\sigma_1 - \sigma_3) [\tau^2 + (\sigma_n - \sigma_1)(\sigma_n - \sigma_3)]$$

$$\Rightarrow n_2^2 = - \frac{\tau^2 + (\sigma_n - \sigma_1)(\sigma_n - \sigma_3)}{(\sigma_2 - \sigma_3)(\sigma_1 - \sigma_2)}$$

$$x_3 = n_3^2 = \frac{\det \underline{A}_3}{\det \underline{A}} \quad \underline{A}_3 = \begin{pmatrix} \sigma_1 & \sigma_2 & \sigma_n \\ \sigma_1^2 & \sigma_2^2 & \sigma_n^2 \\ 1 & 1 & 1 \end{pmatrix}$$

$$\det \underline{A}_3 = -(\sigma_1 - \sigma_2) [\tau^2 + (\sigma_n - \sigma_1)(\sigma_n - \sigma_2)]$$

$$\Rightarrow n_3^2 = \frac{\tau^2 + (\sigma_n - \sigma_1)(\sigma_n - \sigma_2)}{(\sigma_1 - \sigma_3)(\sigma_2 - \sigma_3)}$$

Summary:

$$n_1^2 = \frac{\tau^2 + (\sigma_n - \sigma_2)(\sigma_n - \sigma_3)}{(\sigma_1 - \sigma_2)(\sigma_1 - \sigma_3)}$$

$$n_2^2 = - \frac{\tau^2 + (\sigma_n - \sigma_1)(\sigma_n - \sigma_3)}{(\sigma_2 - \sigma_3)(\sigma_1 - \sigma_2)}$$

$$n_3^2 = \frac{\tau^2 + (\sigma_n - \sigma_1)(\sigma_n - \sigma_2)}{(\sigma_1 - \sigma_3)(\sigma_2 - \sigma_3)}$$

Sanity check: $\sigma_n = \sigma_1$ $\tau = 0$

$$n_1^2 = \frac{0^2 + (\sigma_1 - \sigma_3)(\sigma_1 - \sigma_2)}{(\sigma_1 - \sigma_2)(\sigma_1 - \sigma_3)} = 1 \quad \checkmark$$

$$n_2^2 = (\cancel{\sigma_1 - \sigma_1})(\sigma_1 - \sigma_3) / [(\sigma_2 - \sigma_3)(\sigma_1 - \sigma_2)] = 0$$

$$n_3^2 = (\cancel{\sigma_1 - \sigma_1}) \dots = 0$$

$$\left. \begin{array}{l} n_1^2 = 1 \\ n_2^2 = 0 \\ n_3^2 = 0 \end{array} \right\} \underline{n} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

Non-Negativity constraints: $n_i^2 \geq 0$

denominators are positive

3 inequalities:

$$1) \tau^2 + (\sigma_n - \sigma_2)(\sigma_n - \sigma_3) \geq 0$$

$$2) \tau^2 + (\sigma_n - \sigma_1)(\sigma_n - \sigma_3) \leq 0 \quad (\text{minus sign})$$

$$3) \tau^2 + (\sigma_n - \sigma_1)(\sigma_n - \sigma_2) \geq 0$$

Note: n_i 's have been eliminated

Dependence of σ_n & τ on principal stresses

Completing squares:

$$(\sigma_n - p)(\sigma_n - q) = \left(\sigma_n - \frac{p+q}{2}\right)^2 - \left(\frac{p-q}{2}\right)^2$$

Mohr's circles

Circle 1 ($n_1^2 \geq 0$):

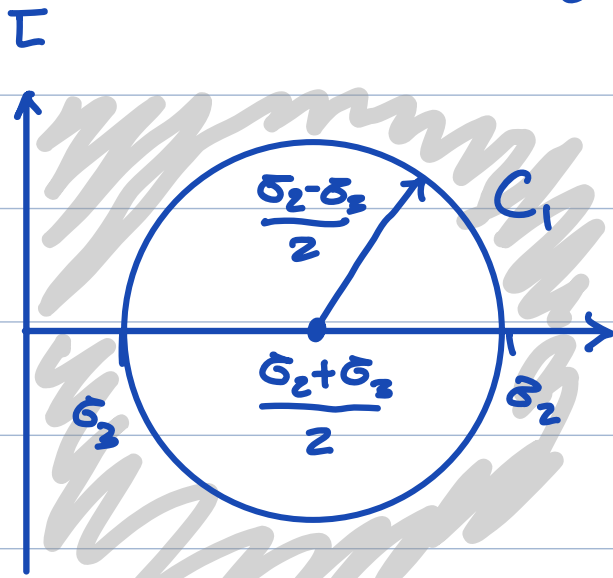
$$\tau^2 + (\sigma_n - \sigma_2)(\sigma_n - \sigma_3) \geq 0$$

$$\tau^2 + \left(\sigma_n - \frac{\sigma_2 + \sigma_3}{2}\right)^2 - \left(\frac{\sigma_2 - \sigma_3}{2}\right)^2 \geq 0$$

$$\tau^2 + \left(\sigma_n - \frac{\sigma_2 + \sigma_3}{2}\right)^2 \geq \left(\frac{\sigma_2 - \sigma_3}{2}\right)^2$$

Eqn of circle: $(x - x_0)^2 + (y - y_0)^2 = R^2$

(x_0, y_0) = center R = radius



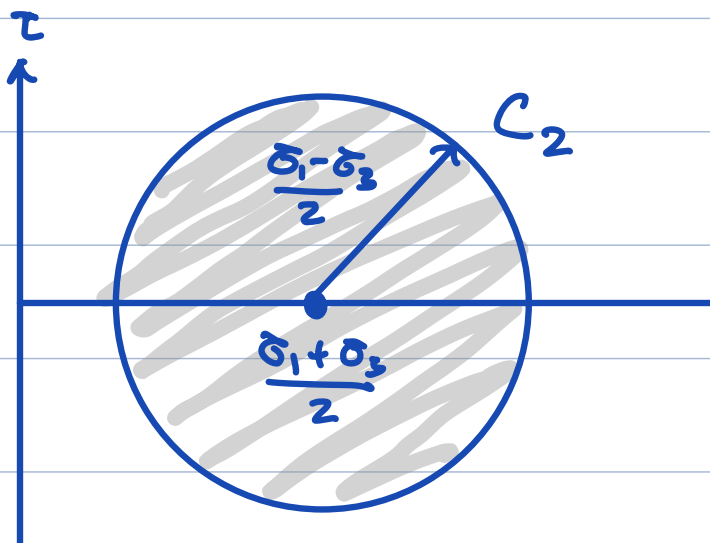
admissible τ & σ_n
are outside C_1

Circle 2 ($n_2^2 \geq 0$):

$$\tau^2 + (\sigma_n - \sigma_1)(\sigma_n - \sigma_3) \leq 0$$

$$\tau^2 + \left(\sigma_n - \frac{\sigma_1 + \sigma_3}{2}\right)^2 - \left(\frac{\sigma_1 - \sigma_3}{2}\right)^2 \leq 0$$

$$\tau^2 + \left(\sigma_n - \frac{\sigma_1 + \sigma_3}{2}\right)^2 \leq \left(\frac{\sigma_1 - \sigma_3}{2}\right)^2$$



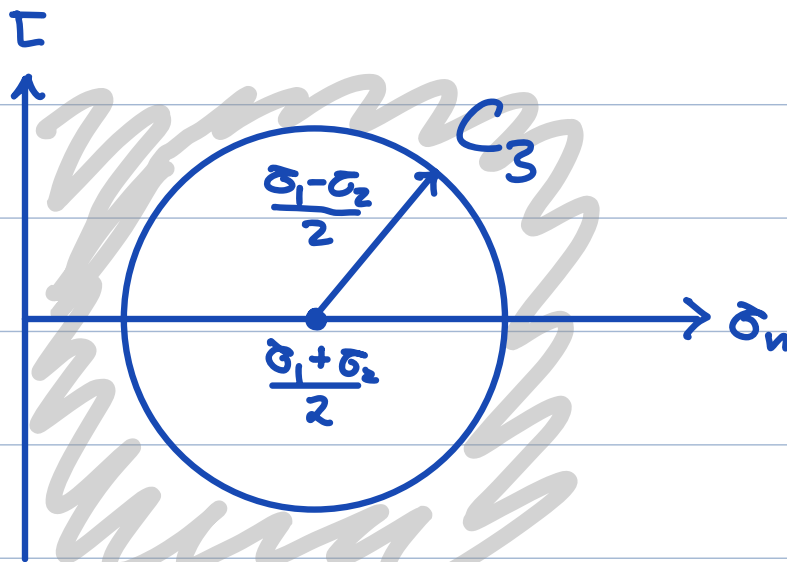
admissible τ & σ_n
are inside circle

Circle 3 ($n_s^2 \geq 0$)

$$\tau^2 + (\sigma_n - \sigma_1)(\sigma_n - \sigma_2) \geq 0$$

$$\tau^2 + \left(\sigma_n - \frac{\sigma_1 + \sigma_2}{2}\right)^2 - \left(\frac{\sigma_1 - \sigma_2}{2}\right)^2 \geq 0$$

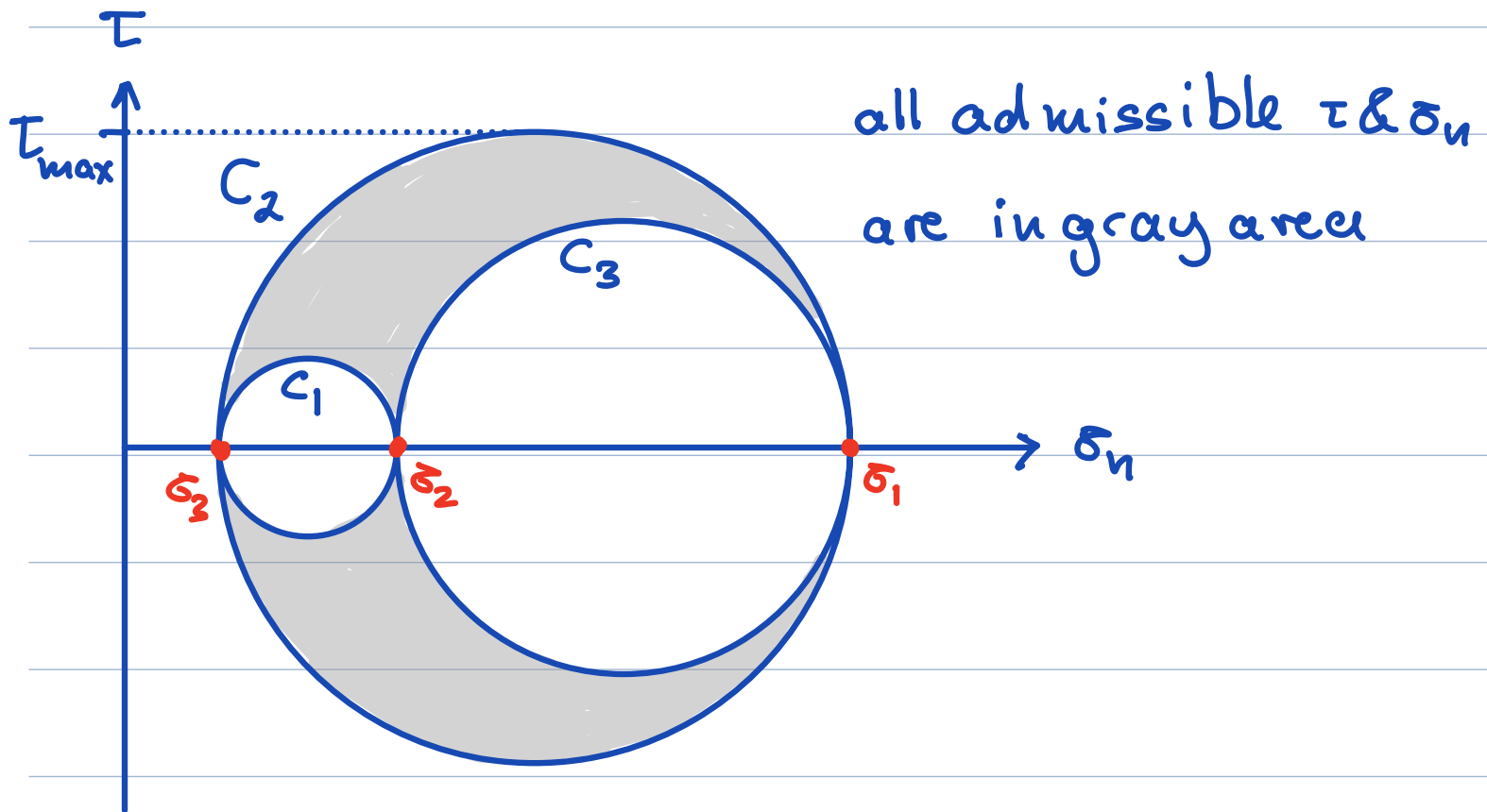
$$\tau^2 + \left(\sigma_n - \frac{\sigma_1 + \sigma_2}{2}\right)^2 \geq \left(\frac{\sigma_1 - \sigma_2}{2}\right)^2$$



admissible τ & σ_n
are outside circle

Combine 3 different conditions
 $\Rightarrow$ 3D Mohr circle

3 D Mohr circle



Sanity check: $\sigma_n = \sigma_i \rightarrow \tau = 0$

no shear stress on principal planes

max shear stress determined by C_2

$$\tau^2 + \left(\sigma_n - \frac{\sigma_1 + \sigma_3}{2} \right)^2 \leq \underbrace{\left(\frac{\sigma_1 - \sigma_3}{2} \right)^2}_{\text{radius}}$$

$$\tau_{max} = \pm \frac{\sigma_1 - \sigma_3}{2}$$

corresponding $\sigma_n = \frac{\sigma_1 + \sigma_3}{2} \neq 0$

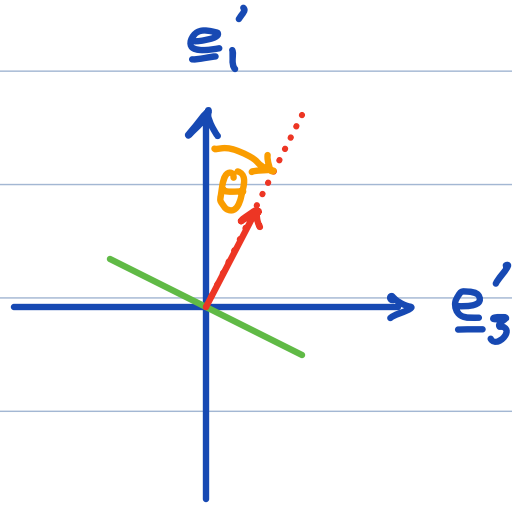
What is the orientation?

$$C_2 \rightarrow n_2^2 \geq 0 \quad \text{on } C_2 \quad n_2 = 0.$$

$$\Rightarrow n_1^2 + n_3^2 = 1$$

$$n_1 = \cos \theta$$

$$n_3 = \sin \theta$$



substitute into σ_n :

$$\sigma_n = \sigma_1 n_1^2 + \sigma_3 n_3^2 = \sigma_1 \cos^2 \theta + \sigma_3 \sin^2 \theta$$

double angle identities: $\cos^2 \theta = \frac{1 + \cos 2\theta}{2}$

$$\sin^2 \theta = \frac{1 - \cos 2\theta}{2}$$

$$\Rightarrow \sigma_n = \underbrace{\frac{\sigma_1 + \sigma_3}{2}}_{\text{center} = C_2} + \underbrace{\frac{\sigma_1 - \sigma_3}{2}}_{\text{radius} = R_2} \cos 2\theta$$

substitute into τ :

$$\tau^2 = \sigma_1^2 n_1^2 + \sigma_2^2 n_2^2 - \sigma_n^2$$

many algebra utilizing $\cos^2 \theta + \sin^2 \theta = 1$

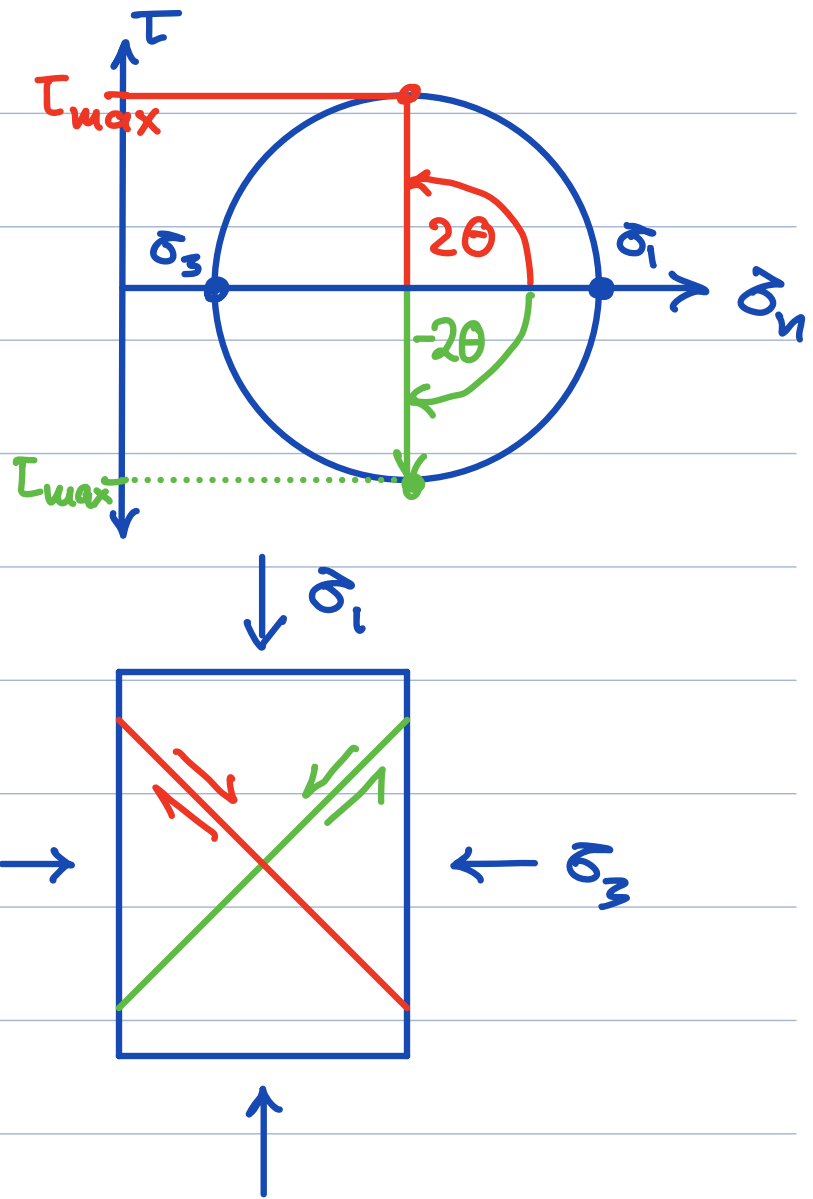
$$\tau = \underbrace{\frac{\sigma_1 - \sigma_3}{2}}_{R_2} \sin 2\theta$$

Summary

$$\sigma_n = C_2 + R_2 \cos(2\theta)$$

$$\tau = R_2 \sin(2\theta)$$

$$\Rightarrow \tau_{\max} \text{ at } \pm 45^\circ \text{ to } \underline{e}_1$$



Do materials fail
on plane with
maximum $|\tau|$?

Empirical criteria for shear failure

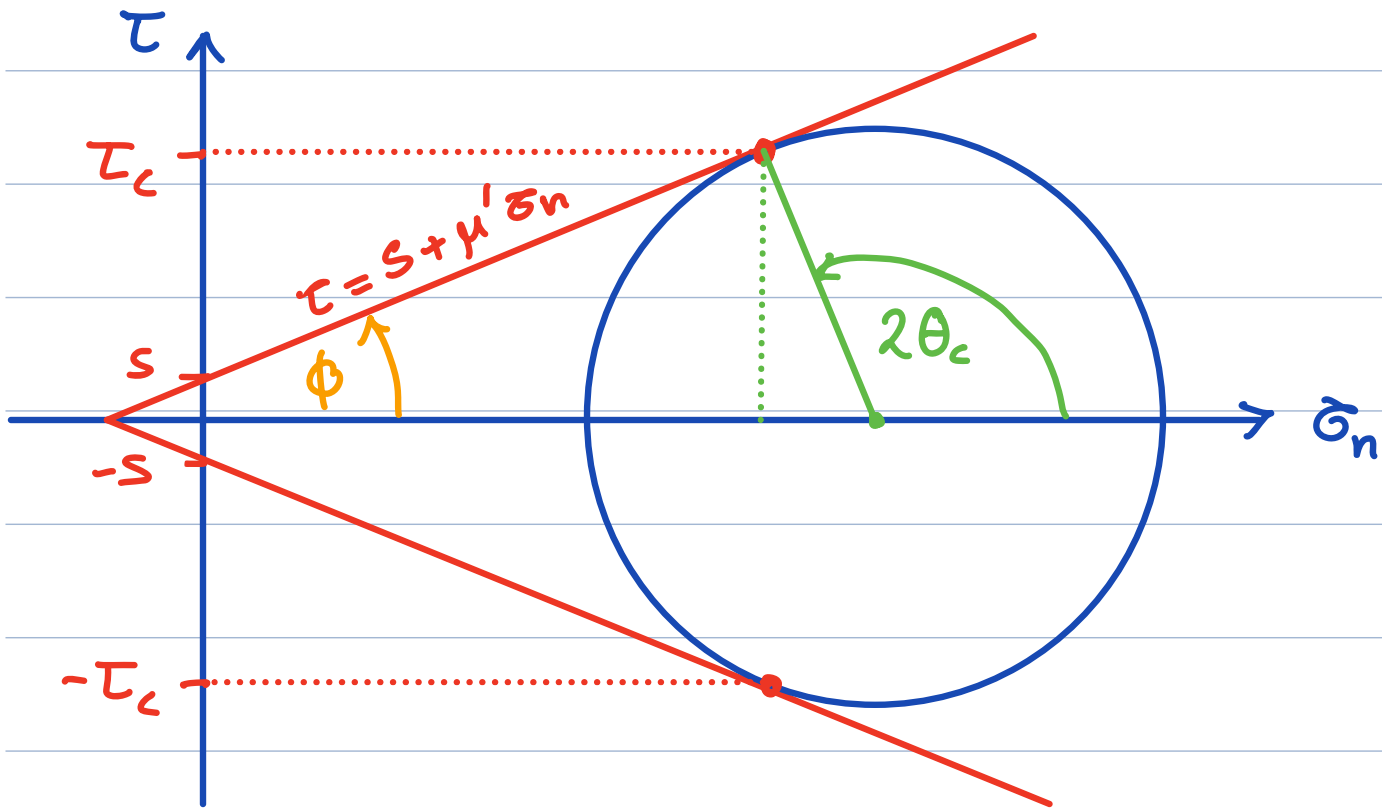
Coulomb failure envelope: $|\tau| = S + \mu' \sigma_n$

S = cohesive strength $\sim 10-100$ MPa

$\mu' = \tan \phi$ internal friction ~ 0.6

$\phi \approx 30^\circ$ angle of internal friction

Mohr - Coulomb Failure

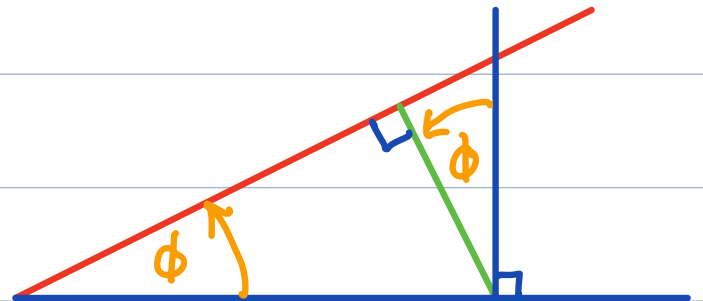


$\Rightarrow$ failure occurs at $\tau_c < \tau_{max}$

Angle of failure?

$$2\theta_c = 90^\circ + \phi$$

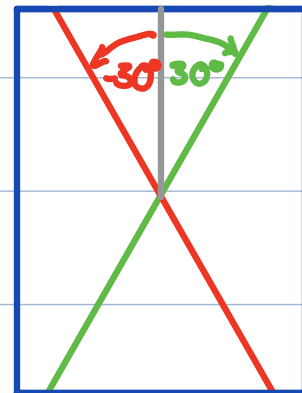
$$\Rightarrow \theta_c = 45^\circ + \frac{\phi}{2}$$



$$\phi \approx 30^\circ$$

$$\theta_c = 45^\circ + 15^\circ = \underline{\underline{60^\circ}}$$

$\Rightarrow$ plane is $\pm 30^\circ$ from e'_1



Here we just use Mohr circle to derive maximum shear stress and Mohr-Coulomb to find failure plane.

There is much more to say:

- GEO 428 Structural Geology
- GEO 322M Basin Geomechanics
- PGE 334 Reservoir Geomechanics